MPG No. 4.3.11 m' [Type III, monoclinic] [M tensor]

* Rank 0 tensor.

$$\begin{bmatrix} M \end{bmatrix}$$

$$M = \mathbb{M}_0^{(1)}$$

* Rank 1 tensor.

$$\begin{bmatrix} M_x & 0 & M_z \end{bmatrix}$$

$$M_x = \mathbb{M}_x^{(1)}$$
$$M_z = \mathbb{M}_z^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} M_{xx} & 0 & M_{xz} \\ 0 & M_{yy} & 0 \\ M_{xz} & 0 & M_{zz} \end{bmatrix}$$

$$M_{xx} = \mathbb{M}_0^{(1)} - \mathbb{M}_u^{(1)} + \mathbb{M}_v^{(1)}$$
$$M_{xz} = \mathbb{M}_{xz}^{(1)}$$
$$M_{yy} = \mathbb{M}_0^{(1)} - \mathbb{M}_u^{(1)} - \mathbb{M}_v^{(1)}$$
$$M_{zz} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_u^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & 0 & M_{xz} \\ 0 & 0 & 0 \\ -M_{xz} & 0 & 0 \end{bmatrix}$$

$$M_{xz} = -\mathbb{T}_y^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} M_{xxx} & 0 & M_{xxz} \\ M_{yyx} & 0 & M_{yyz} \\ M_{zzx} & 0 & M_{zzz} \\ 0 & M_{yyz} & 0 \\ M_{xxz} & 0 & M_{zzx} \\ 0 & M_{yyx} & 0 \end{bmatrix}$$

$$M_{xxx} = 3\mathbb{M}_x^{(1)} + 2\mathbb{M}_x^{\alpha(1)}$$
$$M_{xxz} = \mathbb{M}_z^{(1)} - \mathbb{M}_z^{\alpha(1)} + \mathbb{M}_z^{\beta(1)}$$
$$M_{yyx} = \mathbb{M}_x^{(1)} - \mathbb{M}_x^{\alpha(1)} + \mathbb{M}_x^{\beta(1)}$$
$$M_{yyz} = \mathbb{M}_z^{(1)} - \mathbb{M}_z^{\alpha(1)} - \mathbb{M}_z^{\beta(1)}$$
$$M_{zzx} = \mathbb{M}_x^{(1)} - \mathbb{M}_x^{\alpha(1)} - \mathbb{M}_x^{\beta(1)}$$
$$M_{zzz} = 3\mathbb{M}_z^{(1)} + 2\mathbb{M}_z^{\alpha(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ M_{yyx} & 0 & M_{yyz} \\ M_{zzx} & 0 & 0 \\ 0 & -\frac{M_{yyz}}{2} & 0 \\ -\frac{M_{xxz}}{2} & 0 & -\frac{M_{zzx}}{2} \\ 0 & -\frac{M_{yyx}}{2} & 0 \end{bmatrix}$$

$$M_{xxz} = 2\mathbb{M}_z^{(2)} + 2\mathbb{T}_{xy}^{(1)}$$
$$M_{yyx} = 2\mathbb{M}_x^{(2)} + 2\mathbb{T}_{yz}^{(1)}$$
$$M_{yyz} = 2\mathbb{M}_z^{(2)} - 2\mathbb{T}_{xy}^{(1)}$$
$$M_{zzx} = 2\mathbb{M}_x^{(2)} - 2\mathbb{T}_{yz}^{(1)}$$
$$\frac{M_{yyz}}{2} = \mathbb{M}_z^{(2)} - \mathbb{T}_{xy}^{(1)}$$
$$\frac{M_{xxz}}{2} = \mathbb{M}_z^{(2)} + \mathbb{T}_{xy}^{(1)}$$
$$\frac{M_{zzx}}{2} = \mathbb{M}_x^{(2)} - \mathbb{T}_{yz}^{(1)}$$
$$\frac{M_{yyx}}{2} = \mathbb{M}_x^{(2)} + \mathbb{T}_{yz}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & M_{yzy} & 0 \\ M_{zxx} & 0 & M_{zxz} \\ 0 & M_{xyy} & 0 \end{bmatrix}$$

$$M_{yzy} = \mathbb{M}_z^{(3)} + \mathbb{T}_{xy}^{(2)}$$
$$M_{zxx} = -\mathbb{M}_z^{(3)} + \mathbb{T}_{xy}^{(2)}$$
$$M_{zxz} = \mathbb{M}_x^{(3)} + \mathbb{T}_{yz}^{(2)}$$
$$M_{xyy} = -\mathbb{M}_x^{(3)} + \mathbb{T}_{yz}^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} M_{xxxx} & M_{xxyy} & M_{xxzz} & 0 & M_{xxzx} & 0 \\ M_{xxyy} & M_{yyyy} & M_{yyzz} & 0 & M_{yyzx} & 0 \\ M_{xxzz} & M_{yyzz} & M_{zzzz} & 0 & M_{zzzx} & 0 \\ 0 & 0 & 0 & M_{yzyz} & 0 & M_{yzxy} \\ M_{xxzx} & M_{yyzx} & M_{zzzx} & 0 & M_{zxzx} & 0 \\ 0 & 0 & 0 & M_{yzxy} & 0 & M_{xyxy} \end{bmatrix}$$

$$M_{xxxx} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)} - 4\mathbb{M}_u^{(2)} + 2\mathbb{M}_v^{(1)} + 4\mathbb{M}_v^{(2)}$$
$$M_{xxyy} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)}$$
$$M_{xxzz} = \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} + \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(1)} + \mathbb{M}_v^{(1)}$$
$$M_{xxzx} = -\mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)}$$
$$M_{yyyy} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} + \mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)} - 4\mathbb{M}_u^{(2)} - 2\mathbb{M}_v^{(1)} - 4\mathbb{M}_v^{(2)}$$
$$M_{yyzz} = \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(1)} - \mathbb{M}_v^{(1)}$$
$$M_{yyzx} = 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)}$$
$$M_{zzzz} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_0^{(2)} + 2\mathbb{M}_{4u}^{(1)} + 2\mathbb{M}_4^{(1)} + 4\mathbb{M}_u^{(1)} + 8\mathbb{M}_u^{(2)}$$
$$M_{zzzx} = \mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)}$$
$$M_{yzyz} = \mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(2)} - \mathbb{M}_v^{(2)}$$
$$M_{yzxy} = 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(2)}$$
$$M_{zxzx} = \mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} + \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(2)} + \mathbb{M}_v^{(2)}$$
$$M_{xyxy} = \mathbb{M}_0^{(2)} + 2\mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & M_{xxyy} & M_{xxzz} & 0 & M_{xxzx} & 0 \\ -M_{xxyy} & 0 & M_{yyzz} & 0 & M_{yyzx} & 0 \\ -M_{xxzz} & -M_{yyzz} & 0 & 0 & M_{zzzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{yzxy} \\ -M_{xxzx} & -M_{yyzx} & -M_{zzzx} & 0 & 0 & 0 \\ 0 & 0 & 0 & -M_{yzxy} & 0 & 0 \end{bmatrix}$$

$$
\begin{aligned}
M_{xxyy} &= -2\mathbb{M}_{v}^{(3)} + 4\mathbb{T}_{xyz}^{(1)} \\
M_{xxzz} &= 3\mathbb{M}_{u}^{(3)} - \mathbb{M}_{v}^{(3)} - 4\mathbb{T}_{xyz}^{(1)} \\
M_{xxzx} &= \mathbb{M}_{xz}^{(3)} - 2\mathbb{T}_{y}^{(1)} + 2\mathbb{T}_{y}^{\alpha(1)} + 2\mathbb{T}_{y}^{\beta(1)} \\
M_{yyzz} &= 3\mathbb{M}_{u}^{(3)} + \mathbb{M}_{v}^{(3)} + 4\mathbb{T}_{xyz}^{(1)} \\
M_{yyzx} &= \mathbb{M}_{xz}^{(3)} - 4\mathbb{T}_{y}^{\beta(1)} \\
M_{zzzx} &= \mathbb{M}_{xz}^{(3)} + 2\mathbb{T}_{y}^{(1)} - 2\mathbb{T}_{y}^{\alpha(1)} + 2\mathbb{T}_{y}^{\beta(1)} \\
M_{yzxy} &= \mathbb{T}_{y}^{(1)} + 4\mathbb{T}_{y}^{\alpha(1)}
\end{aligned}
$$

* Rank 4 tensor (aas).

$$
\begin{bmatrix}
M_{yzyz} & 0 & M_{yzxy} \\
0 & M_{zxzx} & 0 \\
M_{yzxy} & 0 & M_{xyxy}
\end{bmatrix}
$$

$$
\begin{aligned}
M_{yzyz} &= \mathbb{M}_{0}^{(3)} - 2\mathbb{M}_{u}^{(4)} + 2\mathbb{M}_{v}^{(4)} \\
M_{yzxy} &= 2\mathbb{M}_{xz}^{(4)} \\
M_{zxzx} &= \mathbb{M}_{0}^{(3)} - 2\mathbb{M}_{u}^{(4)} - 2\mathbb{M}_{v}^{(4)} \\
M_{xyxy} &= \mathbb{M}_{0}^{(3)} + 4\mathbb{M}_{u}^{(4)}
\end{aligned}
$$

* Rank 4 tensor (aaa).

$$
\begin{bmatrix}
0 & 0 & M_{yzxy} \\
0 & 0 & 0 \\
-M_{yzxy} & 0 & 0
\end{bmatrix}
$$

$$
M_{yzxy} = \mathbb{T}_{y}^{(2)}
$$

* Rank 4 tensor (sa).

$$
\begin{bmatrix}
0 & M_{xxzx} & 0 \\
0 & M_{yyzx} & 0 \\
0 & M_{zzzx} & 0 \\
M_{yzyz} & 0 & M_{yzxy} \\
0 & M_{zxzx} & 0 \\
M_{xyyz} & 0 & M_{xyxy}
\end{bmatrix}
$$

$$
\begin{aligned}
M_{xxzx} &= 2\mathbb{M}_{xz}^{(5)} + \mathbb{T}_{y}^{(3)} - \mathbb{T}_{y}^{\alpha(2)} - \mathbb{T}_{y}^{\beta(2)} \\
M_{yyzx} &= \mathbb{T}_{y}^{(3)} + 2\mathbb{T}_{y}^{(4)} + 2\mathbb{T}_{y}^{\alpha(2)} \\
M_{zzzx} &= -2\mathbb{M}_{xz}^{(5)} + \mathbb{T}_{y}^{(3)} - \mathbb{T}_{y}^{\alpha(2)} + \mathbb{T}_{y}^{\beta(2)} \\
M_{yzyz} &= -3\mathbb{M}_{u}^{(5)} - \mathbb{M}_{v}^{(5)} + \mathbb{T}_{xyz}^{(2)} \\
M_{yzxy} &= \mathbb{M}_{xz}^{(5)} + \mathbb{T}_{y}^{(4)} - \mathbb{T}_{y}^{\alpha(2)} + \mathbb{T}_{y}^{\beta(2)} \\
M_{zxzx} &= 3\mathbb{M}_{u}^{(5)} - \mathbb{M}_{v}^{(5)} + \mathbb{T}_{xyz}^{(2)} \\
M_{xyyz} &= -\mathbb{M}_{xz}^{(5)} + \mathbb{T}_{y}^{(4)} - \mathbb{T}_{y}^{\alpha(2)} - \mathbb{T}_{y}^{\beta(2)} \\
M_{xyxy} &= 2\mathbb{M}_{v}^{(5)} + \mathbb{T}_{xyz}^{(2)}
\end{aligned}
$$

* Rank 4 tensor (as).

$$
\begin{bmatrix}
0 & 0 & 0 & M_{yzyz} & 0 & M_{yzxy} \\
M_{zxxx} & M_{zxyy} & M_{zxzz} & 0 & M_{zxzx} & 0 \\
0 & 0 & 0 & M_{xyyz} & 0 & M_{xyxy}
\end{bmatrix}
$$

$$
\begin{aligned}
M_{yzyz} &= -3\mathbb{M}_{u}^{(6)} - \mathbb{M}_{v}^{(6)} + \mathbb{T}_{xyz}^{(3)} \\
M_{yzxy} &= -\mathbb{M}_{xz}^{(6)} + \mathbb{T}_{y}^{(6)} - \mathbb{T}_{y}^{\alpha(3)} - \mathbb{T}_{y}^{\beta(3)} \\
M_{zxxx} &= 2\mathbb{M}_{xz}^{(6)} + \mathbb{T}_{y}^{(5)} - \mathbb{T}_{y}^{\alpha(3)} - \mathbb{T}_{y}^{\beta(3)} \\
M_{zxyy} &= \mathbb{T}_{y}^{(5)} + 2\mathbb{T}_{y}^{(6)} + 2\mathbb{T}_{y}^{\alpha(3)} \\
M_{zxzz} &= -2\mathbb{M}_{xz}^{(6)} + \mathbb{T}_{y}^{(5)} - \mathbb{T}_{y}^{\alpha(3)} + \mathbb{T}_{y}^{\beta(3)} \\
M_{zxzx} &= 3\mathbb{M}_{u}^{(6)} - \mathbb{M}_{v}^{(6)} + \mathbb{T}_{xyz}^{(3)} \\
M_{xyyz} &= \mathbb{M}_{xz}^{(6)} + \mathbb{T}_{y}^{(6)} - \mathbb{T}_{y}^{\alpha(3)} + \mathbb{T}_{y}^{\beta(3)} \\
M_{xyxy} &= 2\mathbb{M}_{v}^{(6)} + \mathbb{T}_{xyz}^{(3)}
\end{aligned}
$$

* Rank 4 tensor (s).

$$
\begin{bmatrix}
M_{xxxx} & M_{xxyy} & M_{xxzz} & 0 & M_{xxzx} & 0 & 0 & M_{xxxz} & 0 \\
M_{yyxx} & M_{yyyy} & M_{yyzz} & 0 & M_{yyzx} & 0 & 0 & M_{yyxz} & 0 \\
M_{zzxx} & M_{zzyy} & M_{zzzz} & 0 & M_{zzzx} & 0 & 0 & M_{zzxz} & 0 \\
0 & 0 & 0 & M_{yzyz} & 0 & M_{yzxy} & M_{yzzy} & 0 & M_{yzyx} \\
M_{zxxx} & M_{zxyy} & M_{zxzz} & 0 & M_{zxzx} & 0 & 0 & M_{zxxz} & 0 \\
0 & 0 & 0 & M_{xyyz} & 0 & M_{xyxy} & M_{xyzy} & 0 & M_{xyyx}
\end{bmatrix}
$$

$$M_{xxxx} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)} - 4\mathbb{M}_u^{(2)} + 2\mathbb{M}_v^{(1)} + 4\mathbb{M}_v^{(2)}$$

$$M_{xxyy} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)} - 2\mathbb{M}_v^{(3)} + 4\mathbb{T}_{xyz}^{(1)}$$

$$M_{xxzz} = \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} + \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(1)} + 3\mathbb{M}_u^{(3)} + \mathbb{M}_v^{(1)} - \mathbb{M}_v^{(3)} - 4\mathbb{T}_{xyz}^{(1)}$$

$$M_{xxzx} = -\mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} + 2\mathbb{M}_{xz}^{(5)} - 2\mathbb{T}_y^{(1)} + \mathbb{T}_y^{(3)} + 2\mathbb{T}_y^{\alpha(1)} - \mathbb{T}_y^{\alpha(2)} + 2\mathbb{T}_y^{\beta(1)} - \mathbb{T}_y^{\beta(2)}$$

$$M_{xxxz} = -\mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} - 2\mathbb{M}_{xz}^{(5)} - 2\mathbb{T}_y^{(1)} - \mathbb{T}_y^{(3)} + 2\mathbb{T}_y^{\alpha(1)} + \mathbb{T}_y^{\alpha(2)} + 2\mathbb{T}_y^{\beta(1)} + \mathbb{T}_y^{\beta(2)}$$

$$M_{yyxx} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)} + 2\mathbb{M}_v^{(3)} - 4\mathbb{T}_{xyz}^{(1)}$$

$$M_{yyyy} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} + \mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)} - 4\mathbb{M}_u^{(2)} - 2\mathbb{M}_v^{(1)} - 4\mathbb{M}_v^{(2)}$$

$$M_{yyzz} = \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(1)} + 3\mathbb{M}_u^{(3)} - \mathbb{M}_v^{(1)} + \mathbb{M}_v^{(3)} + 4\mathbb{T}_{xyz}^{(1)}$$

$$M_{yyzx} = 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + \mathbb{M}_{xz}^{(3)} + \mathbb{T}_y^{(3)} + 2\mathbb{T}_y^{(4)} + 2\mathbb{T}_y^{\alpha(2)} - 4\mathbb{T}_y^{\beta(1)}$$

$$M_{yyxz} = 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + \mathbb{M}_{xz}^{(3)} - \mathbb{T}_y^{(3)} - 2\mathbb{T}_y^{(4)} - 2\mathbb{T}_y^{\alpha(2)} - 4\mathbb{T}_y^{\beta(1)}$$

$$M_{zzxx} = \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} + \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(1)} - 3\mathbb{M}_u^{(3)} + \mathbb{M}_v^{(1)} + \mathbb{M}_v^{(3)} + 4\mathbb{T}_{xyz}^{(1)}$$

$$M_{zzyy} = \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(1)} - 3\mathbb{M}_u^{(3)} - \mathbb{M}_v^{(1)} - \mathbb{M}_v^{(3)} - 4\mathbb{T}_{xyz}^{(1)}$$

$$M_{zzzz} = \mathbb{M}_0^{(1)} + 2\mathbb{M}_0^{(2)} + 2\mathbb{M}_{4u}^{(1)} + 2\mathbb{M}_4^{(1)} + 4\mathbb{M}_u^{(1)} + 8\mathbb{M}_u^{(2)}$$

$$M_{zzzx} = \mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} - 2\mathbb{M}_{xz}^{(5)} + 2\mathbb{T}_y^{(1)} + \mathbb{T}_y^{(3)} - 2\mathbb{T}_y^{\alpha(1)} - \mathbb{T}_y^{\alpha(2)} + 2\mathbb{T}_y^{\beta(1)} + \mathbb{T}_y^{\beta(2)}$$

$$M_{zzxz} = \mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} + 2\mathbb{M}_{xz}^{(5)} + 2\mathbb{T}_y^{(1)} - \mathbb{T}_y^{(3)} - 2\mathbb{T}_y^{\alpha(1)} + \mathbb{T}_y^{\alpha(2)} + 2\mathbb{T}_y^{\beta(1)} - \mathbb{T}_y^{\beta(2)}$$

$$M_{yzyz} = \mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(2)} - 3\mathbb{M}_u^{(5)} - \mathbb{M}_v^{(2)} - \mathbb{M}_v^{(5)} + \mathbb{T}_{xyz}^{(2)}$$

$$M_{yzxy} = 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(5)} + \mathbb{T}_y^{(1)} + \mathbb{T}_y^{(4)} + 4\mathbb{T}_y^{\alpha(1)} - \mathbb{T}_y^{\alpha(2)} + \mathbb{T}_y^{\beta(2)}$$

$$M_{yzzy} = \mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(2)} + 3\mathbb{M}_u^{(5)} - \mathbb{M}_v^{(2)} + \mathbb{M}_v^{(5)} - \mathbb{T}_{xyz}^{(2)}$$

$$M_{yzyx} = 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(5)} + \mathbb{T}_y^{(1)} - \mathbb{T}_y^{(4)} + 4\mathbb{T}_y^{\alpha(1)} + \mathbb{T}_y^{\alpha(2)} - \mathbb{T}_y^{\beta(2)}$$

$$M_{zxxx} = -\mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(3)} + 2\mathbb{T}_y^{(1)} - 2\mathbb{T}_y^{\alpha(1)} - 2\mathbb{T}_y^{\beta(1)}$$

$$M_{zxyy} = 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} - \mathbb{M}_{xz}^{(3)} + 4\mathbb{T}_y^{\beta(1)}$$

$$M_{zxzz} = \mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(3)} - 2\mathbb{T}_y^{(1)} + 2\mathbb{T}_y^{\alpha(1)} - 2\mathbb{T}_y^{\beta(1)}$$

$$M_{zxzx} = \mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} + \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(2)} + 3\mathbb{M}_u^{(5)} + \mathbb{M}_v^{(2)} - \mathbb{M}_v^{(5)} + \mathbb{T}_{xyz}^{(2)}$$

$$M_{zxxz} = \mathbb{M}_0^{(2)} - \mathbb{M}_{4u}^{(1)} + \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(2)} - 3\mathbb{M}_u^{(5)} + \mathbb{M}_v^{(2)} + \mathbb{M}_v^{(5)} - \mathbb{T}_{xyz}^{(2)}$$

$$M_{xyyz} = 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(5)} - \mathbb{T}_y^{(1)} + \mathbb{T}_y^{(4)} - 4\mathbb{T}_y^{\alpha(1)} - \mathbb{T}_y^{\alpha(2)} - \mathbb{T}_y^{\beta(2)}$$

$$M_{xyxy} = \mathbb{M}_0^{(2)} + 2\mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(2)} + 2\mathbb{M}_v^{(5)} + \mathbb{T}_{xyz}^{(2)}$$

$$M_{xyzy} = 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(5)} - \mathbb{T}_y^{(1)} - \mathbb{T}_y^{(4)} - 4\mathbb{T}_y^{\alpha(1)} + \mathbb{T}_y^{\alpha(2)} + \mathbb{T}_y^{\beta(2)}$$

$$M_{xyyx} = \mathbb{M}_0^{(2)} + 2\mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(2)} - 2\mathbb{M}_v^{(5)} - \mathbb{T}_{xyz}^{(2)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & 0 & M_{yzxy} & M_{yzzy} & 0 & M_{yzyx} \\ M_{zxxx} & M_{zxyy} & M_{zxzz} & 0 & M_{zxzx} & 0 & 0 & M_{zxxz} & 0 \\ 0 & 0 & 0 & M_{xyyz} & 0 & M_{xyxy} & M_{xyzy} & 0 & M_{xyyx} \end{bmatrix}$$

$$
\begin{aligned}
M_{yzyz} &= \mathbb{M}_0^{(3)} - 2\mathbb{M}_u^{(4)} - 3\mathbb{M}_u^{(6)} + 2\mathbb{M}_v^{(4)} - \mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)} \\
M_{yzxy} &= 2\mathbb{M}_{xz}^{(4)} - \mathbb{M}_{xz}^{(6)} + \mathbb{T}_y^{(2)} + \mathbb{T}_y^{(6)} - \mathbb{T}_y^{\alpha(3)} - \mathbb{T}_y^{\beta(3)} \\
M_{yzzy} &= -\mathbb{M}_0^{(3)} + 2\mathbb{M}_u^{(4)} - 3\mathbb{M}_u^{(6)} - 2\mathbb{M}_v^{(4)} - \mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)} \\
M_{yzyx} &= -2\mathbb{M}_{xz}^{(4)} - \mathbb{M}_{xz}^{(6)} - \mathbb{T}_y^{(2)} + \mathbb{T}_y^{(6)} - \mathbb{T}_y^{\alpha(3)} - \mathbb{T}_y^{\beta(3)} \\
M_{zxxx} &= 2\mathbb{M}_{xz}^{(6)} + \mathbb{T}_y^{(5)} - \mathbb{T}_y^{\alpha(3)} - \mathbb{T}_y^{\beta(3)} \\
M_{zxyy} &= \mathbb{T}_y^{(5)} + 2\mathbb{T}_y^{(6)} + 2\mathbb{T}_y^{\alpha(3)} \\
M_{zxzz} &= -2\mathbb{M}_{xz}^{(6)} + \mathbb{T}_y^{(5)} - \mathbb{T}_y^{\alpha(3)} + \mathbb{T}_y^{\beta(3)} \\
M_{zxzx} &= \mathbb{M}_0^{(3)} - 2\mathbb{M}_u^{(4)} + 3\mathbb{M}_u^{(6)} - 2\mathbb{M}_v^{(4)} - \mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)} \\
M_{zxxz} &= -\mathbb{M}_0^{(3)} + 2\mathbb{M}_u^{(4)} + 3\mathbb{M}_u^{(6)} + 2\mathbb{M}_v^{(4)} - \mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)} \\
M_{xyyz} &= 2\mathbb{M}_{xz}^{(4)} + \mathbb{M}_{xz}^{(6)} - \mathbb{T}_y^{(2)} + \mathbb{T}_y^{(6)} - \mathbb{T}_y^{\alpha(3)} + \mathbb{T}_y^{\beta(3)} \\
M_{xyxy} &= \mathbb{M}_0^{(3)} + 4\mathbb{M}_u^{(4)} + 2\mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)} \\
M_{xyzy} &= -2\mathbb{M}_{xz}^{(4)} + \mathbb{M}_{xz}^{(6)} + \mathbb{T}_y^{(2)} + \mathbb{T}_y^{(6)} - \mathbb{T}_y^{\alpha(3)} + \mathbb{T}_y^{\beta(3)} \\
M_{xyyx} &= -\mathbb{M}_0^{(3)} - 4\mathbb{M}_u^{(4)} + 2\mathbb{M}_v^{(6)} + \mathbb{T}_{xyz}^{(3)}
\end{aligned}
$$

* Rank 4 tensor (t).

$$
\begin{bmatrix}
M_{xxxx} & 0 & M_{xxxz} \\
0 & M_{yyyy} & 0 \\
M_{zzzx} & 0 & M_{zzzz} \\
M_{yyzx} & 0 & M_{yyzz} \\
M_{zzxx} & 0 & M_{zzxz} \\
0 & M_{xxyy} & 0 \\
0 & M_{yzzy} & 0 \\
M_{zxxx} & 0 & M_{zxxz} \\
M_{xxyy} + M_{yyzz} - M_{yzzy} - M_{zxxz} + M_{zzxx} & 0 & M_{xyyz} \\
0 & M_{xyyz} + \frac{M_{zzxz}}{2} - \frac{M_{zzzx}}{2} & 0
\end{bmatrix}
$$

$$
\begin{aligned}
M_{xxxx} &= 3\mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)} - 4\mathbb{M}_u^{(2)} + 2\mathbb{M}_v^{(1)} + 4\mathbb{M}_v^{(2)} \\
M_{xxxz} &= -\mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} - 2\mathbb{M}_{xz}^{(5)} \\
M_{yyyy} &= 3\mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} + \mathbb{M}_{4v}^{(1)} + 2\mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)} - 4\mathbb{M}_u^{(2)} - 2\mathbb{M}_v^{(1)} - 4\mathbb{M}_v^{(2)} \\
M_{zzzx} &= \mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} - 2\mathbb{M}_{xz}^{(5)} \\
M_{zzzz} &= 3\mathbb{M}_0^{(1)} + 2\mathbb{M}_{4u}^{(1)} + 2\mathbb{M}_4^{(1)} + 4\mathbb{M}_u^{(1)} + 8\mathbb{M}_u^{(2)} \\
M_{yyzx} &= 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + \mathbb{M}_{xz}^{(3)} \\
M_{yyzz} &= \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(1)} + 3\mathbb{M}_u^{(3)} - \mathbb{M}_v^{(1)} + \mathbb{M}_v^{(3)} \\
M_{zzxx} &= \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} + \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(1)} - 3\mathbb{M}_u^{(3)} + \mathbb{M}_v^{(1)} + \mathbb{M}_v^{(3)} \\
M_{zzxz} &= \mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} + \mathbb{M}_{xz}^{(3)} + 2\mathbb{M}_{xz}^{(5)} \\
M_{xxyy} &= \mathbb{M}_0^{(1)} + 2\mathbb{M}_{4u}^{(1)} - \mathbb{M}_4^{(1)} - 2\mathbb{M}_u^{(1)} - 2\mathbb{M}_v^{(3)} \\
M_{yzzy} &= \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} - \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(2)} + 3\mathbb{M}_u^{(5)} - \mathbb{M}_v^{(2)} + \mathbb{M}_v^{(5)} \\
M_{zxxx} &= -\mathbb{M}_{4y}^{\alpha(1)} - \mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(1)} + 2\mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(3)} \\
M_{zxxz} &= \mathbb{M}_0^{(1)} - \mathbb{M}_{4u}^{(1)} + \mathbb{M}_{4v}^{(1)} - \mathbb{M}_4^{(1)} + \mathbb{M}_u^{(2)} - 3\mathbb{M}_u^{(5)} + \mathbb{M}_v^{(2)} + \mathbb{M}_v^{(5)} \\
M_{xyyz} &= 2\mathbb{M}_{4y}^{\beta(1)} + \mathbb{M}_{xz}^{(2)} - \mathbb{M}_{xz}^{(5)}
\end{aligned}
$$